Language-Conditioned QAOA for Portfolio Selection: A Bias-Controlled Study Reveals a Training-Horizon Confound

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Abstract. We introduce *language-conditioned* QAOA: a pipeline in which a large language model (LLM) reads financial news and emits per-asset convictions that tilt the cost Hamiltonian of a cardinality-constrained Quantum Approximate Optimization Algorithm (QAOA) portfolio optimizer. We then ask, with strong bias controls, whether the language adds out-of-sample value. It appears to: an LLM released during the evaluation window (qwen2.5-14B, Sept. 2024) beats matched random tilts by +0.76%/quarter on changed-optimum instances. But a *genuinely out-of-sample* LLM (Llama 2, cutoff Sept. 2022) on the *same* headlines yields −0.23% ($p = 0.94$, indistinguishable from random): the benefit appears only for the model released into the test period—consistent with *training-horizon hindsight* rather than prediction (the pre-cutoff model is also smaller, a caveat we note). We contribute the language→quantum conditioning pipeline, a reusable bias-control template, and a quantified training-horizon look-ahead warning for LLM-for-finance evaluation.

Keywords: QAOA · Quantum optimization · LLM-conditioning · Look-ahead bias · Quantum finance.

1 Pipeline and Method

Two lines straddle a gap: Lee et al. [3] drive Black–Litterman views with an LLM but optimize *classically*, while Chen et al. [1] bring numerically predicted views into a *quantum* pipeline with no natural language. We occupy the middle—an LLM's language conditions the quantum optimizer—and treat "do news views help?" as a *measurement problem*. Holding B of n assets to maximize the mean–variance objective $\mu^\top x - q\, x^\top \Sigma x$ over a fixed Dow-30 universe (current, not point-in-time, constituents; CRSP returns; shrinkage covariance), we map this binary selection to a diagonal Ising Hamiltonian solved by QAOA [2]. A local LLM reads RavenPack news headlines time-stamped *strictly before* each quarterly decision date into convictions $v_i \in [-1, 1]$; a ±10% view tilts $\mu \to \mu + s\, v$ (the standard sentiment-to-view convention), estimated on the train window and evaluated on the held-out next quarter. Bias controls: hundreds of (subset × window) instances from a fixed seed, pre-registered metrics, a matched-random-tilt *placebo*, an off-the-shelf-sentiment (RavenPack ESS) ablation, and—decisively—two LLMs on the *same* headlines: **qwen2.5-14B** (released Sept. 2024, training

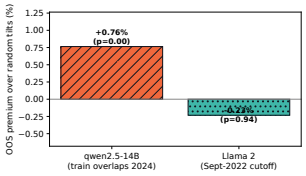


Fig. 1. Out-of-sample premium over matched random tilts: a training-overlapping LLM (qwen2.5-14B) vs. a pre-cutoff LLM (Llama 2) on identical headlines.

plausibly *spans* the window) and **Llama 2** (documented pretraining cutoff Sept. 2022, *before* it). The language effect is read at the exact optimum, free of optimizer noise.

2 Results

Over 1,600 instances (Dow 30, n=14 qubits, hold 5, four quarterly dates 2023-12–2024-09), the training-overlapping LLM beats matched random tilts by a language-specific premium of +0.76%/quarter on changed-optimum instances (instance-level $p \approx 0$, positive in 3/4 quarters); the pre-cutoff LLM, on identical headlines, yields −0.23% ($p = 0.94$, 2/4)—indistinguishable from random (Fig. 1). The conditioning has real mechanism leverage (a ±10% view changes the exact optimum in 47–68% of instances), yet QAOA at depth p=4 recovers the true optimum only 7.7% of the time; we make no quantum-advantage claim. Limitations: four decision dates (instance-level p is pseudo-replicated); the clean and contaminated LLMs differ in capability as well as cutoff, so the two are not fully separated; small-n simulation; fixed-membership large-cap US; no costs. None overturn the takeaway: an LLM whose training overlaps the backtest can manufacture out-of-sample "alpha" that a pre-cutoff model does not—a look-ahead that LLM-for-finance studies must guard against by dating their model against the evaluation window. Fixed seeds; code released upon publication.

References

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2. Farhi, E., Goldstone, J., Gutmann, S.: A quantum approximate optimization algorithm. arXiv preprint arXiv:1411.4028 (2014)
3. Lee, Y., Kim, Y., Kim, J., Kim, S., Lee, Y.: LLM-enhanced Black-Litterman portfolio optimization. In: CIKM Workshop on Financial AI (2025)

Does Language Really Help a Quantum Portfolio Optimizer?

A Bias-Controlled Study and a Training-Horizon Confound in LLM-Conditioned QAOA

QNLPAI 2026 Student Poster Session · author details withheld for double-blind review

The Central Question

When a backtest says "LLM news-views help a quantum portfolio optimizer," is that **signal — or an artifact?** We build a deliberately bias-controlled pipeline to catch ourselves.

Experimental Pipeline

01

Universe Selection

Dow 30, CRSP total returns — objective, fixed universe

02

LLM Conviction Generation

LLM reads RavenPack headlines time-stamped **strictly before** each decision date → per-asset ±10% view tilts

03

QAOA Cost Hamiltonian

Conviction tilt applied to QAOA cost function → hold 5 of 14 assets, evaluate held-out next quarter

04

Controls Applied

Matched random-tilt placebo; ablation vs. RavenPack sentiment; **two LLMs on identical headlines**

The Two-LLM Design

qwen2.5-14B

Training overlaps 2024 — **contaminated**

Llama 2

Sept-2022 cutoff — genuinely **out-of-sample**

Both models read the *same* RavenPack headlines. Only the training-horizon relationship to the backtest period differs.

⚠ **Key Finding — The apparent "language signal" is the LLM's hindsight:** The out-of-sample alpha collapses to indistinguishable-from-random with a genuinely out-of-sample model.

+0.76% per quarter

qwen2.5-14B Premium

Over random tilts (p≈0) — training overlaps 2024 backtest window

−0.23% per quarter

Llama 2 Premium

Over random tilts (p=0.94) — Sept-2022 cutoff, indistinguishable from random

47–68%

Optima Changed

A ±10% view changes the exact optimum in 47–68% of instances — language has real mechanism leverage on the selection.

7.7%

True Optimum Found

Shallow QAOA is a weak exact optimizer — no quantum-advantage claim is made

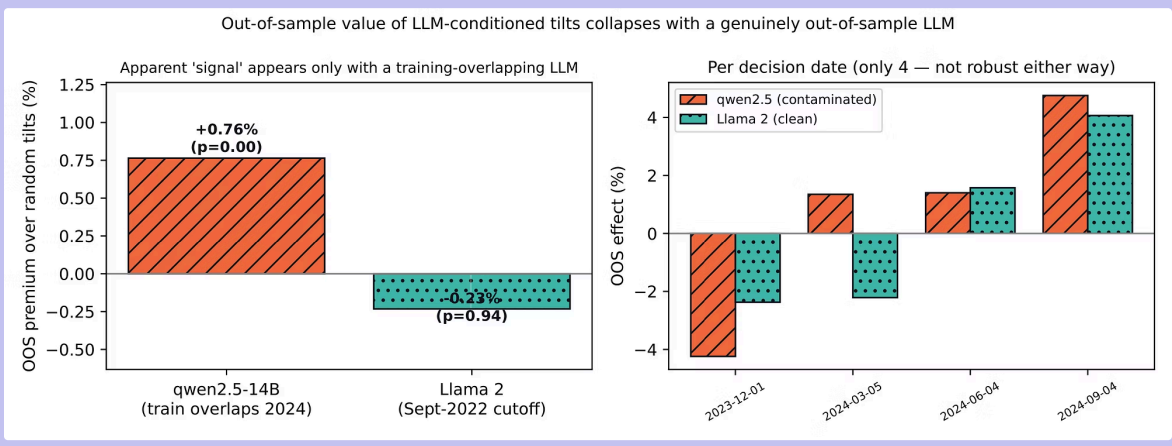


FIGURE 1 — Out-of-sample value of LLM-conditioned tilts collapses with a genuinely out-of-sample LLM.

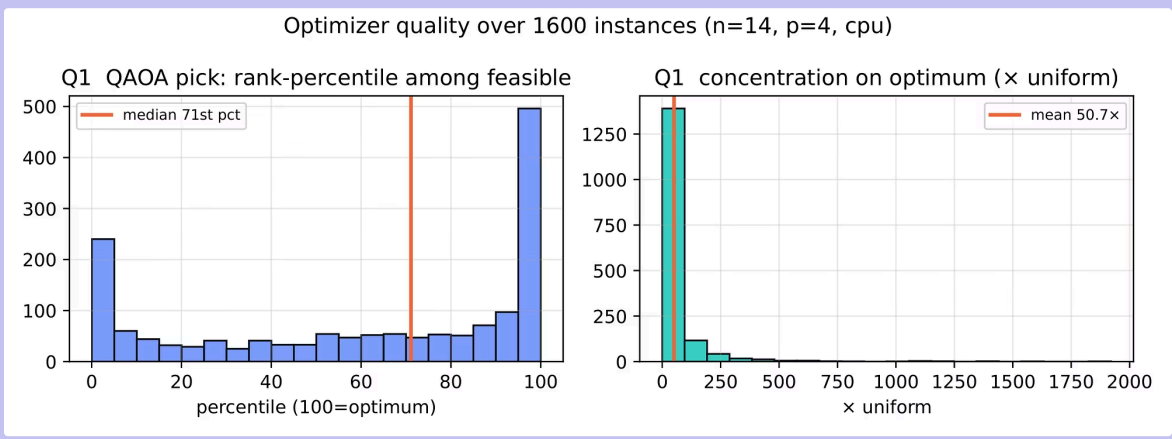


FIGURE 2 — QAOA optimizer quality over 1600 instances (n=14, p=4): rank-percentile among feasible portfolios (median 71st) and concentration on the optimum (mean 50.7× uniform).

Honest Limitations

- Only **4 decision dates** — instance-level p is pseudo-replicated
- Clean and contaminated LLMs differ in **capability** as well as cutoff date
- Small-n **exact simulation**; large-cap US equities only; no transaction costs
- None of these limitations **inflate** the central result

Contribution

→ A **reproducible bias-control template** for LLM-conditioned financial backtests

→ A **quantified warning**: date your LLM against your backtest window, or training-horizon look-ahead will manufacture alpha that isn't there

✓ **Reproducibility:** Fixed seeds, committed CRSP + conviction caches; code released upon publication.